



Experiment 8

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Branch: CSE

Semester: 6th

Subject Name: Full Stack Development – II

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Section/Group: KRG 3-B

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Subject Code: 23CSH-309

1. Aim: To design and implement a Spring Boot application that integrates third-party APIs including **Stripe Payment Gateway** for handling online transactions and **Gmail SMTP Service** for sending email notifications after successful payments.

2. Objective:

- To understand the concept of third-party API integration
- To integrate Stripe Payment Gateway for secure online payments
- To configure and use Gmail SMTP for sending emails
- To implement REST APIs for payment processing
- To handle payment success and failure responses
- To send automated email notifications after successful payment
- To follow Layered Architecture (Controller – Service – Repository)
- To enhance real-world application functionality using external services

3. Implementation / Code:

▪ Tools & Technologies Used:-

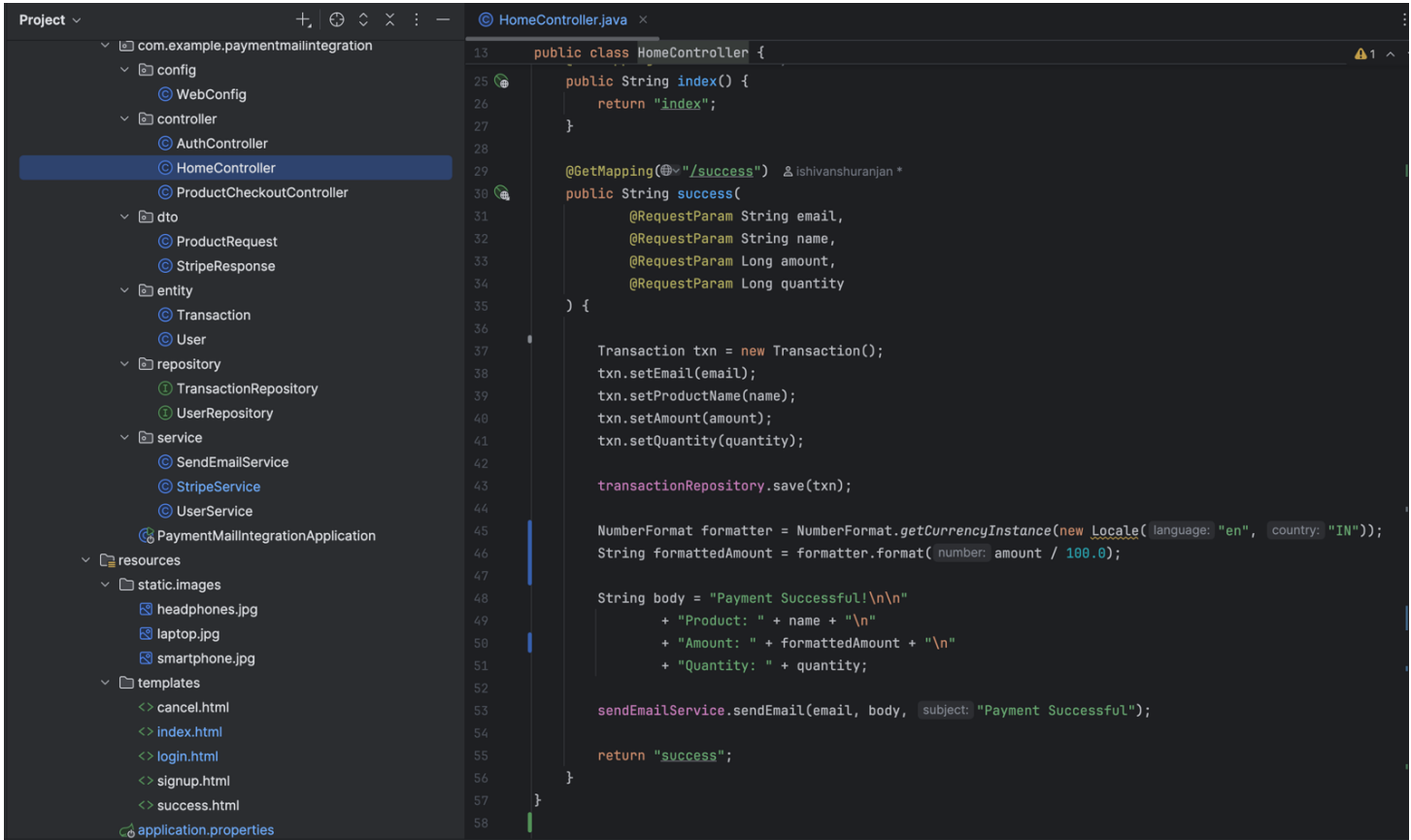
- Java
- Spring Boot
- Spring Web
- Spring Data JPA
- Maven
- IntelliJ IDEA
- Stripe API
- Gmail SMTP (Java Mail Sender)
- Postman / Browser
- PostgreSQL Database

▪ Implementation Description:-

- A Spring Boot project named Payment Mail Integration is created.
- The project integrates Stripe API for handling online payments securely.
- Users enter card details on Stripe Checkout page and complete payment.

- After successful payment, a success page is displayed.
- A confirmation email is automatically sent to the user using Gmail SMTP.

■ Project Structure:-



The screenshot shows an IDE with a project structure on the left and the `HomeController.java` file open in the editor. The project structure includes:

- `com.example.paymentmailintegration`
 - `config`
 - `WebConfig`
 - `controller`
 - `AuthController`
 - `HomeController` (selected)
 - `ProductCheckoutController`
 - `dto`
 - `ProductRequest`
 - `StripeResponse`
 - `entity`
 - `Transaction`
 - `User`
 - `repository`
 - `TransactionRepository`
 - `UserRepository`
 - `service`
 - `SendEmailService`
 - `StripeService`
 - `UserService`
 - `PaymentMailIntegrationApplication`
 - `resources`
 - `static.images`
 - `headphones.jpg`
 - `laptop.jpg`
 - `smartphone.jpg`
 - `templates`
 - `cancel.html`
 - `index.html`
 - `login.html`
 - `signup.html`
 - `success.html`
 - `application.properties`

The `HomeController.java` file contains the following code:

```
13 public class HomeController {
25     public String index() {
26         return "index";
27     }
28
29     @GetMapping("/success") & ishivanshuranjan *
30     public String success(
31         @RequestParam String email,
32         @RequestParam String name,
33         @RequestParam Long amount,
34         @RequestParam Long quantity
35     ) {
36
37         Transaction txn = new Transaction();
38         txn.setEmail(email);
39         txn.setProductName(name);
40         txn.setAmount(amount);
41         txn.setQuantity(quantity);
42
43         transactionRepository.save(txn);
44
45         NumberFormat formatter = NumberFormat.getCurrencyInstance(new Locale("en", "IN"));
46         String formattedAmount = formatter.format(number: amount / 100.0);
47
48         String body = "Payment Successful!\n\n"
49             + "Product: " + name + "\n"
50             + "Amount: " + formattedAmount + "\n"
51             + "Quantity: " + quantity;
52
53         sendEmailService.sendEmail(email, body, subject: "Payment Successful");
54
55         return "success";
56     }
57 }
58
```

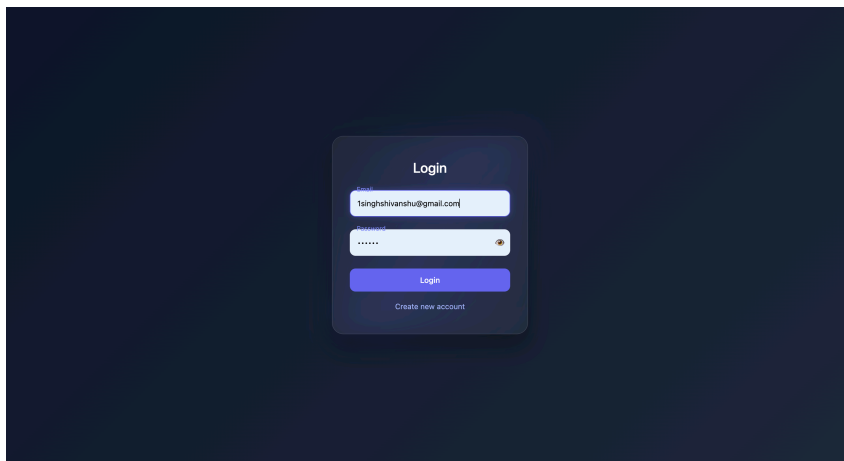
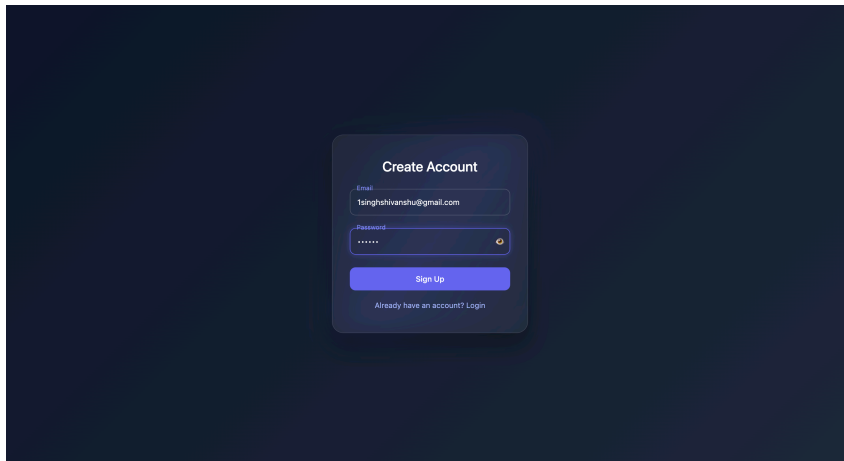
4. Output:

- Spring Boot application started successfully on **localhost:8080**
- Stripe payment page loaded successfully
- Payment completed successfully using test card
- Success page displayed after payment completion
- Email notification sent successfully to registered email
- Email contains product details, amount, and quantity
- Payment flow tested successfully using Stripe test environment
- Application integrated successfully with third-party APIs



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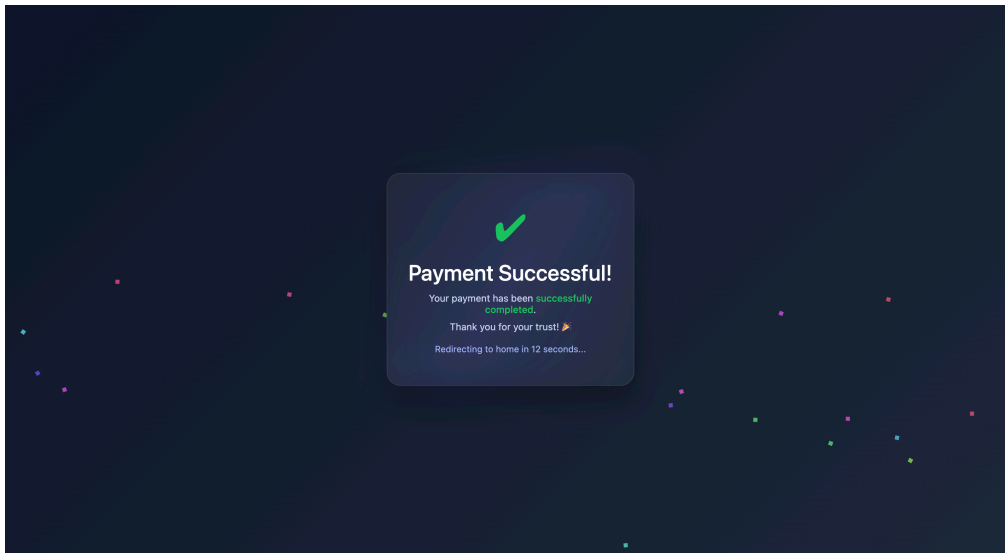
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Payment Successful Inbox x



shivanshusingh0020@gmail.com

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Payment Successful!

Product: Headphones

Amount: ₹50,000.00

Quantity: 2

↩ Reply

➡ Forward



5. Learning Outcomes (What I Have Learnt):

- Understanding of third-party API integration
- Working with Stripe Payment Gateway
- Secure handling of online transactions
- Sending emails using Gmail SMTP
- Implementing real-world payment systems
- Integration of frontend and backend systems
- Usage of REST APIs in real applications
- Handling success and failure scenarios
- Improving application functionality using external services